

Rain Gardens

A Practical Primer for Small Yards

By Tomas Whitcombe

1. Why a garden should drink

When rain falls on a roof, a driveway or a compacted lawn, very little of it soaks in. It runs off, gathering speed and dirt, and is carried through gutters and drains to the nearest stream. On a single lot the volume seems small. Across a neighbourhood it adds up to sudden floods after every storm and streams that run nearly dry between them.

A rain garden reverses that pattern on a small scale. It is a shallow planted basin placed where runoff naturally collects, built so that water pools for a few hours and then sinks into the ground. The soil filters out sediment and many pollutants, the plants take up nutrients, and the water that remains recharges the ground slowly instead of rushing away.

None of this needs special equipment. A typical home rain garden can be dug by hand over a weekend, planted with hardy native species and left largely alone. This primer walks through how one works, how to size it for your roof and soil, and how to keep it healthy once it is in the ground.

The most important idea is that a rain garden is not a pond. It should hold standing water for no more than a day after a heavy storm. If it stays wet longer than that, the basin is too deep, the soil drains too slowly or too much water is being sent to it. Each of those problems has a simple fix, covered in the sections that follow.

2. Anatomy of a rain garden

Figure 1 shows a cross-section through a basin fed by a roof downspout. Water enters at the low side, spreads across a level floor and pools to a depth of six to twelve inches before any excess spills out over a lowered section of the berm.

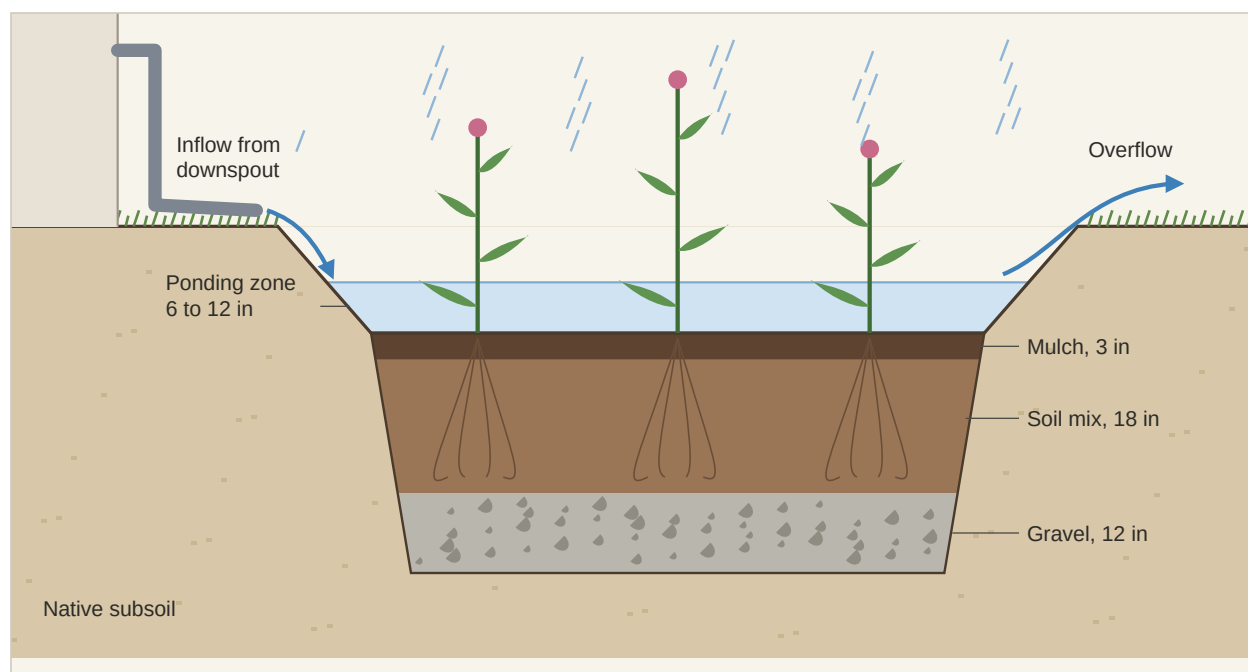


Figure 1. A rain garden in section. Roof water enters from the left, pools above the mulch and soaks down through the soil mix and gravel; anything beyond the ponding depth leaves by the overflow.

Below the surface, three layers do most of the work. A thin blanket of shredded mulch keeps the soil from crusting and holds moisture for the plants during dry spells. Beneath it, a loose mix of sand, topsoil and compost lets water move downward quickly while still trapping fine particles.

At the bottom, a layer of washed gravel stores water temporarily and spreads it across the native subsoil. In heavy clay this layer is what keeps the basin from behaving like a bathtub, giving the water somewhere to wait while it seeps slowly into the ground below.

3. Sizing the basin

Start by estimating the area that drains to the garden. For a downspout, this is the share of the roof that feeds it, measured flat as if seen from above. A simple rule for sandy soils is to make the basin about one fifth of that drainage area. Heavier soils need more room, as the table below suggests.

Test your soil before you dig. Make a hole about a foot deep, fill it with water and let it drain, then fill it again and time how long the second filling takes to disappear. If the level drops an inch or more an hour, the site is suitable for a rain garden. Slower soils can still work with a deeper gravel layer and a larger footprint.

Soil	Drains per hour	Basin size	Gravel depth
Sand	2 in or more	20% of area	None to 6 in
Sandy loam	1 to 2 in	25% of area	6 in
Silt loam	0.5 to 1 in	35% of area	12 in
Clay	Under 0.5 in	45% or more	18 in

Table 1. Starting points for basin size and gravel depth by soil.

Keep the basin at least ten feet from the house foundation and away from septic fields, and never place it over buried utilities. A gentle slope of less than twelve percent is ideal. On steeper ground, a series of small terraced basins is safer than one large excavation.

4. Planting and care

Choose plants that tolerate both wet feet and dry spells, because a rain garden swings between the two. Deep rooted native grasses, sedges and flowering perennials are the backbone of most designs. Place the species that handle standing water best on the basin floor and the more drought tolerant ones along the sloping sides and the berm.

Plant densely, roughly one plant for every square foot, so that the canopy closes over the soil within two seasons and crowds out weeds. Water new plantings weekly during their first summer unless rain does the job for you.

After the first year, care is light. Pull weeds in spring, top up the mulch where it has thinned, and clear the inlet and the overflow of leaves after autumn storms. Once established, a well built rain garden asks for an afternoon or two of work each year and returns cleaner water, fewer floods and a patch of habitat that hums with pollinators all summer.

A year of care, at a glance

- Spring: pull weeds and top up thin mulch.
- Summer: water new plants weekly in the first year.
- Autumn: clear leaves from the inlet and overflow.
- After big storms: check the basin drains within a day.